Research Article

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Investigating trade-offs made by American football linebackers using tracking data

https://doi.org/10.1515/jqas-2022-0091
Received October 26, 2022; accepted June 13, 2023;
published online July 21, 2023

Abstract: In recent years, the game of football has made a shift towards being more quantitative. With the advent of charting and tracking data, player evaluation is able to be studied from several different angles. In this paper, we build and refine two novel metrics: Bite Distance Under Expected (BDUE) and Ground Covered Over Expected (GCOE) for the evaluation of linebackers in the National Football League (NFL). Here, we show that these metrics are heavily correlated with each other, which demonstrates the trade-off linebackers have to make between being aggressive against the run and being effective when the opposing offense is using play-action. We also show that these metrics are more stable than those in the public space. Finally, we show how these metrics measure deception by opposing offenses.

Keywords: American football; machine learning; tracking data; trade-offs

1 Introduction

The game of American football (hereafter referred to as football) has undergone a major shift towards being more quantitative. Many of these changes are finding their way into public-facing media such as newspapers, radio shows and television sets. Most public-facing uses of football data are focused on the decision-making processes of teams, either off of the field (Clark 2019; Rodrigue 2021) or on the field (Baldwin 2022; Burke 2013), and a pimary example is in the form of fourth-down decisions. However, through a variety of new data sources (Football Outsiders 2022; Next Gen Stats 2022; PFF 2022), player evaluation has improved drastically from past attempts to quantify the game, utilizing public play-by-play data like expected points added (EPA), charting data from Pro Football Focus (PFF 2022) and other companies, and most recently x- and y-coordinate tracking data (Bliss 2022; Eager et al. 2022; Lopez 2020; Next Gen Stats 2022).

In football there are generally two types of plays; run plays and pass plays. Historically run plays have been the most heavily-used plays due to the relative safety of trying to move the ball while carrying it. The famous Ohio State coach Woody Hayes once said “only three things can happen when you pass (a completion, an incompletion, and an interception) and two of them are bad” (Life Magazine 1969). Linebackers, and other players playing in the “box” (players that are lined up near the line of scrimmage and between the opposing offense’s left and right tackles – or end players on the line of scrimmage) are players tasked with stopping both the run and the pass, and are often the target of deception by opposing offenses. The rise of passing as the dominant way to move the football has included an increase in plays that are called play-action passes, which are passing plays in which the offense fakes a handoff to a potential ballcarrier before dropping back to pass. These plays are very successful, and have been so even through an increase in their usage (Baldwin 2018a; Baldwin 2018b; Eager et al. 2022; Hermsmeyer 2019). As play-action usage has increased, our understanding of the linebacker position has not necessarily grown with it, due in large part to the trade-offs that are necessary for them to be proficient in defending both the run and the pass.

In Eager et al. (2022) we introduced three new metrics that combined the charting data from PFF and the x and y coordinate tracking data from the National Football League (NFL) in an effort to improve upon player evaluation. One of these metrics, Bite Distance Under Expected (BDUE), measured how much linebackers and other box players reacted to play-action passes during the 2017 to 2020 NFL seasons. Measuring how much defenders react to such a play (whether they “bite” or not) gives us an edge in evaluating defensive players as well as offensive schemes and their play callers.

While not biting on play-action passes is ostensibly a good thing, we conjecture it comes at the expense of

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something else. A player who chooses to stay and not bite on play-action passes does that at the expense of defending run plays. This might be rational, as most running plays are not as valuable as passing plays, but in other cases it might be an underreaction. Given the power of both charting and tracking data, we can test this hypothesis.

Thus, in this paper, in addition to making updates to our BDUE model which we use on play-action plays, we build a second model for run plays called Ground Covered Over Expected (GCOE). Specifically, BDUE and GCOE builds on previous work in the football analytics space using tracking data, which has mostly focused on the dropback passing game (Burke 2019; Chu et al. 2020; Deshpande and Evans 2020; Reyers and Swartz 2023). We find that, like BDUE, GCOE is a stable metric from season-to-season on a player-level and, as theorized, is negatively correlated with BDUE. For most players there is a trade-off between being aggressive stopping the run and being discerning on play-action passes. We show that players who buck this trend are players who are able to stop the run effectively and avoid vacating valuable areas of the field on play action. We also show how well offenses induce or prevent linebacker movement, which gives us insight into teams, players and play calling schemes that take advantage of defenses in the NFL.

2 Data

The two main sources of data in this paper are PFF play-by-play charting data (PFF) and the NFL's Next Gen Stats (NGS) tracking data (Next Gen Stats 2022). We use additional data, like stadium and weather data, along with NFL Scouting combine (and college pro data). For a reference on the differences between charting and tracking we recommend a thorough review by Statsbomb (Burriel and Buldu 2021). In summary, charting data is generated by humans' tracking events – features like whether or not a play was a play-action pass – while tracking data is data generated by devises like RFID chips in players' shoulder pads or through computer vision.

2.1 PFF charting data

PFF's charting data feeds feature both a blend of subjective and objective features that can be included in the modeling process for BDUE and GCOE. We provide justification for including these features in the model below:

- **Outside Zone Percentage:** The percent of runs that are labeled as "Outside Zone" by PFF for the offense entering that week on the season. There is evidence to suggest that play action affects defenses differently given a team's scheme (Hasan 2002).
- **Inside Zone Percentage:** The percent of runs that are labeled as "Inside Zone" by PFF for the offense entering that week on the season. There is evidence to suggest that play action affects defenses differently given a team's scheme (Hasan 2002).
- **Power/Counter/Man Percentage:** The percentage of runs that are labeled as "Power", "Counter" or "Man" by PFF for the offense entering that week on the season. There is evidence to suggest that play action affects defenses differently given a team's scheme (Hasan 2002).
- **Down:** The down the play occurred on (1st down, 2nd down, 3rd down or 4th down). Teams run and run play action at different rates on different downs (Football Outsiders 2019).
- **Distance:** The yards left for the offense to get a first down. Teams run and run play action at different rates depending on the yards needed to obtain a first down (Football Outsiders 2019).
- **Quarter:** The quarter of the game the play occurred in (quarters 1–4 in regulation with overtime labeled as quarter 5). Game state affects run and play-action rates (Football Outsiders 2019).
- **Seconds Left in Quarter:** The amount of time, in seconds, left until the quarter finishes. Game state affects run and play-action rates. Game state affects run and play-action rates (Football Outsiders 2019).
- **Offensive and Defensive Score Before:** The amount of points the offense and defense have before the play. Game state affects run and play-action rates. Game state affects run and play-action rates (Football Outsiders 2019).
- **Blitz:** Indicator for if the defense blitzed or not. Blitzes change the composition of coverage behind it.
- **Position:** PFF's charted position of a player on that play based on where they lined up. Player role likely changes player assignment, so we want to account for that.
- **Box Players:** The number of players inside the tackle's width that are at the first or second level of the defense to start the play. The composition of the defense changes the player's roles.
- **Dropback Type and Depth:** The type of dropback (standard, play-action boot, etc.) and how far deep the dropback went. How far the quarterback drops back affects how far the linebackers move on the play.
- **Coverage Scheme:** The category of coverage that defense ran (Cover 1, Cover 2 man, etc.). Different coverage schemes ask different things of box players.

- **Shotgun, Pistol, Motion:** Indicators for whether the offense lined up in shotgun or pistol or used motion. Offensive formation and pre-snap motion affect defensive scheme.
- **Dome:** An indicator for if the roof type of the stadium was a dome. Players move differently in different environments.
- **Turf:** An indicator for if the game was occuring on grass or turf. Players move differently in different environments.
- **Rest:** The amount of days the defensive player had between games. Rest affects player movement and both team's scheme due to differences in preparation time.
- **Run Direction:** The direction of the run (left, middle or right). Different run directions will affect different players and their roles differently.
- **Rusher Position:** The position the rusher is listed as on the roster (RB, FB, etc.). Different rushers will affect players and their roles differently.
- **RPO:** Whether or not the play had a run-pass option concept as part of it. Different offensive schemes will affect defensive players and their roles differently.

2.2 NGS tracking data

Next Gen Stats has a series of data points that are collected through an RFID chip in the player's shoulder pads that relays information every tenth of a second. We provide some justification along with each feature. For the sake of this paper, we do not include the game-level fatigue of the player for a couple of reasons, in large part because it confounds with how good the player is (good players play more snaps than less good players), and we are trying to build an expectation that is independent of how good the player is at football. Additionally, due to the nature of linebacker play – wherein players are generally facing the line of scrimmage and stationary at the snap (when they aren't blitzing), we did not include speed and orientation at the snap:

- **Relative x Coordinate:** Using the ball as the (0, 0) coordinate, this is how far a player is from the ball horizontally (sideline to sideline) at the time of snap. This is a proxy for a player's role.
- **Relative y Coordinate:** Using the ball as the (0, 0) coordinate, this was how far a player is from the ball vertically (endzone to endzone) at the time of the snap. This is a proxy for a player's role.
- **Ball carrier x:** The ball carrier's relative x coordinate at the time frame. This is a proxy for the offensive player's role.
- **Ball carrier y:** The ball carrier's relative y coordinate at the time frame. This is a proxy for the offensive player's role.
- **Ground Covered:** The amount of yards a player covered from their original position at the time of the snap to 2 s of a run play using the difference Euclidean distance between the player and the ballcarrier after 2 s. This is the response variable for the E in GCOE.
- **Bite Distance:** The amount of yards a player moved in the y direction 2 s into a play-action play. This is the reponse variable for the E in BDUE. Negative distance (e.g. displacement) is towards the opponent's line of scrimmage (considered bad), while positive distance is away from the opponent's line of scrimmage. This is the response variable for the E in BDUE.

3 Methods

As highlighted in Section 2, we calculated a player's ground covered and bite distance on each play to get an expected output based on the features described above after adjusting for confounders. For both metrics we used cross validation on a week-by-week level, and then we used root mean squared error to evaluate the predictions. We explored a Null Model (taking the mean of the metric), Elastic-Net Regression, Random Forest, and a tuned XGBoost based on maximum depth and eta (XGBoost 2022). The XGBoost model was tuned using a hyper grid of maximum depth between three and six trees with intervals of one tree and eta between 0.2 and 0.3 with intervals of 0.01, using a five-fold cross validation technique to minimize the RMSE. Once the best iteration was found, the model was trained using the best maximum depth and eta and applied to the data using the week-by-week cross validation discussed earlier.

3.1 Ground covered over expected (GCOE)

Tackling ball carriers is one of a box player's primary jobs. Historically, it was unquestionably their most important jobs, as defenses put an emphasis on stopping the run. This is likely why play action is so effective today and is deployed increasingly frequently in both the National Football League and at the college level (Baldwin 2018a). It also could be why offenses are using more "run-pass options" (Lee 2021), as a quarterback can decide which option he wants to deploy depending on how the box players react to the run fake.

While each of the features in Sections 2.1 and 2.2 influence the expected ground covered and bite distance for box players, we highlight a few of the relationships between them and response variables here.

There is evidence to suggest that teams that deploy certain run concepts (especially outside zone) more frequently are more effective when they call play action passing concepts (Hasan 2002), and hence have a different effect on linebacker movement in the running and play-action game (for a summary of the various run concepts that football teams run, see Palazzolo (2018)). Thus we include the percentage that a team runs outside zone, inside zone and power/counter/man run concepts leading up to that game. We also include the particular

run concept used on a given play, as different run concepts impact the ground covered by linebackers (see Figure 1). Run plays labeled as "trick" cause linebackers to flow with the run game the least while the "man" blocking run concept allows them to flow the most.

Additionally, the PFF charting features include run direction, the position of the ball carrier, how many box players were deployed by the defense, whether the run was based off an RPO, along with indicators for shotgun, pistol and motion. The position of the rusher's impact on how much ground was covered by linebackers is evident in Figure 2.

Lastly, we use the season the play was in, the down and distance, the quarter, the time left in the quarter, the score of the game, whether the offense was the home team in the game, and the roof and turf type of the stadium.

All categorical variables were one-hot encoded. The computer programming language R (R Core Team 2021) was used to implement the models. Twenty-two-fold cross validation (for the 18 weeks of the regular season and four weeks of the playoffs) was used because of temporal dependencies, where performance in one week may be influenced by previous weeks. We avoid data leakage by using cross-validation this way. The results of the different modeling frameworks are shown in Table 1.

While all of the models were directionally similar, in that the year-to-year correlations and the season-by-season leaderboards were roughly the same, XGBoost performed the best, so we used it to create the expected ground covered predictions.

Figure 3 shows the feature importance of the XGBoost model using the built in importance feature the R package provides (R Core Team 2021). There were 28 variables used in the modeling process that became 56 total features due to one hot encoding. The top 20 features were shown in Figure 3 for simplicity. A player's x coordinate at the time of the snap – e.g. how far horizontally from the ball – is the most important variable, followed by the ball carrier's y coordinate – how deep they were in the backfield at the snap. Other positional variables – both for the linebacker and the ball carrier – are also important features, as are the run concept both the opposing offense preferred in the past and ran in the present, which were previewed in Figures 1 and 2.

Once we calculate expected ground covered for each player, we calculate ground covered over expected (GCOE) by taking the difference between actual ground covered and expected ground covered. In Section 4.1 we further analyze GCOE for stability and its interplay with BDUE (Section 4.3). Figure 4 includes a pre- and post-snap example of two players from a 2017 game and their respective ground covered estimates.

3.2 Bite distance under expected (BDUE)

Bite Distance Under Expected (BDUE), which was first developed in Eager et al. (2022) and also studied by Hermsmeyer (2019), has been updated to adjust for more confounders. Along with rushing the passer, coverage is the second part of stopping the passing game. We showed

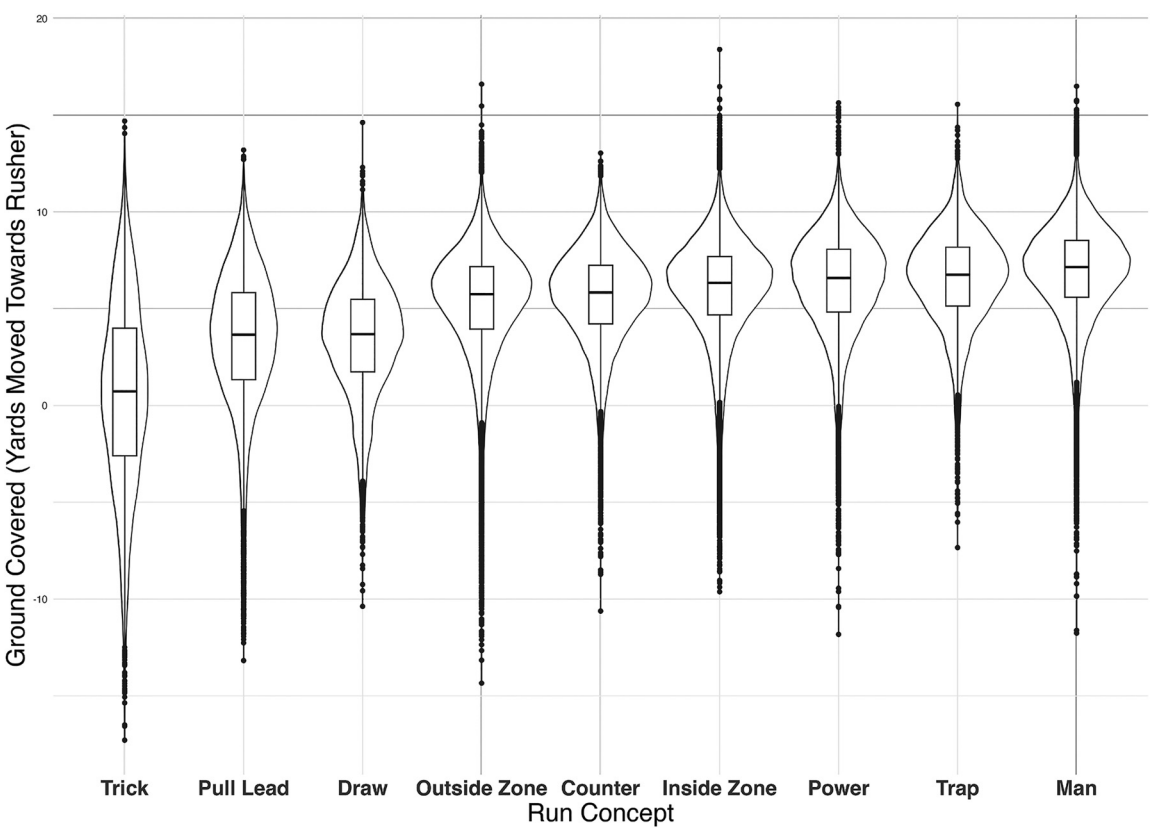


Figure 1: Ground covered by box players as a function of run concept used by the offense. Negative values indicate getting further away from runner from the start of the snap.

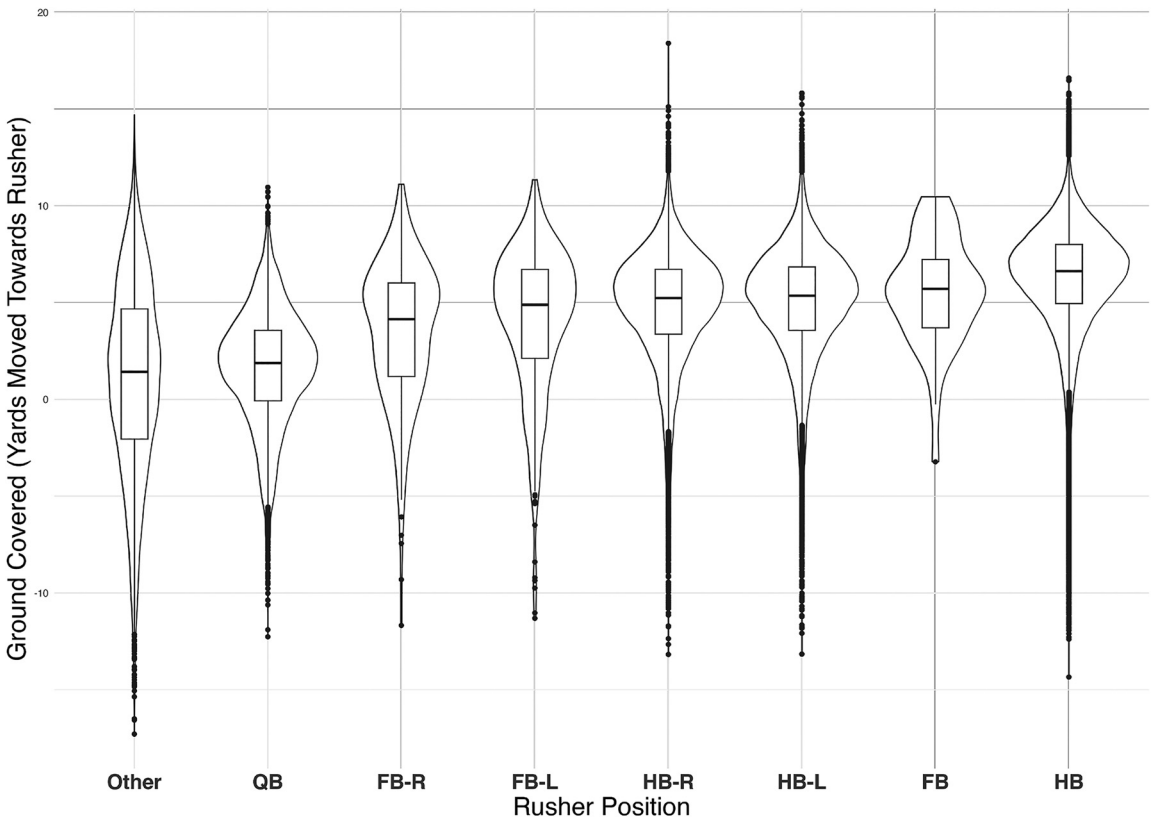


Figure 2: Ground covered by box players as a function of position of the ball carrier. Negative values indicate getting further away from the runner from the start of the snap.

Table 1: RMSE values for different proposed modeling frameworks for GCOE.

Model	RMSE
Null model	3.76 yards
Elastic-net regression	3.15 yards
Random forest	2.47 yards
XGBoost tuned	2.47 yards

in Eager and Chahrouri (2019) that coverage performance, as measured by PFF grades, is both highly correlated with team defensive success in a season and predictive of said success in the following seasons.

However, one criticism of PFF data is that the coverage grades are not nearly as stable year-to-year as other grades and metrics. Another criticism is the grades do not align with conventional opinion – either media consensus or player earnings data. Some of this instability might be due to the nature of coverage, as it is dependent on what the opposing offense is doing. By building BDUE, we aim to reduce some of the noise.

Bite distance is measured as the number of yards, endzone-to-endzone, a box player moved from when the ball was snapped to where they were 2 s into the play. We chose 2 s the measuring point as that is typically the time it takes the quarterback to fake the handoff to the rusher and set his feet to find a receiver (although other time thresholds were tested, the results were qualitatively the same). Similar to GCOE, we adjust for confounders: the rate at which the opposing offense ran each run concept coming into the game, the player's x and y coordinate, game situation information (down, distance, etc.), whether there was a blitz (charted by PFF), the player's position on the play (ILB, FS, etc.), the quarterback's dropback depth and type, the coverage scheme and whether or not it was played in a dome and on turf.

As shown in Figure 5, the position label that a player received on a play based on their alignment affected their bite distance as strong safeties usually had more distance of moving up while middle and inside linebackers usually moved backwards to defend the route concepts.

The player's position on a play in the PFF charting data is an important feature. We only considered players designated as "MLB" (middle linebacker), "LLB" (left linebacker), "RLB" (right linebacker), "LILB" (left inside linebacker), "RILB" (right inside linebacker), "SS" (strong safety), "SSL" (strong safety left) and "SSR" (strong safety right). Keeping these designations allows us to will keep it broad enough to get a good sample size on all the players we wanted to analyze, although it is likely some of these position labels might not always necessarily play in the box (strong safeties for example).

A quarterback's dropback depth in yards – as shown in Figure 6 – influenced how much defenders bite as 7, 8 and 9 step dropbacks usually make defenders bite more while 5 and 6 step dropbacks influenced

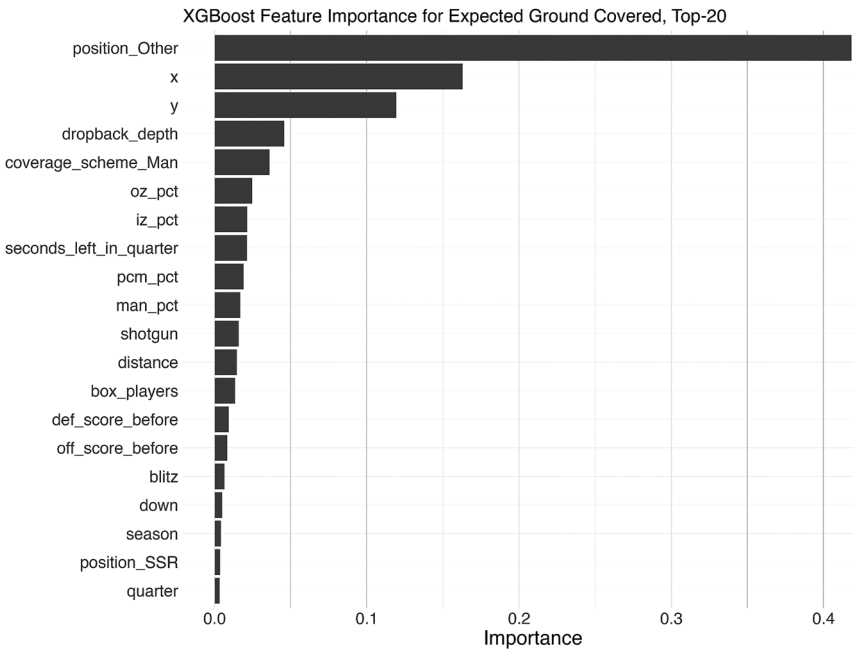
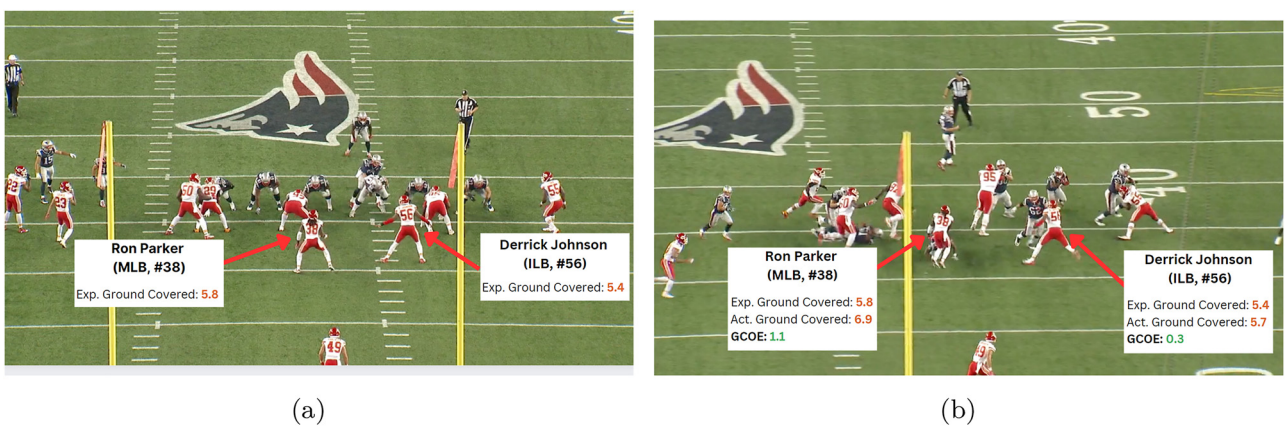


Figure 3: Variable importances for final GCOE model. Only the top 20 features (of 28) are shown.



(a) (b)

Figure 4: Example ground covered over expected instances from a 2017 game between Kansas City and New England. The figure (a) is pre-snap and (b) is post snap.

it less. We evaluate multiple models constructed using these features, using 22-fold cross validation and RMSE, as before. All the models were directionally similar but the Random Forest model performed the best. Its feature importance is shown in Table 2.

For expected bite distance, there were 27 variables with 54 total features due to one hot encoding with the top 20 features being shown in Figure 7 for simplicity. The "vip" package was used on the random forest making the scale different from the feature importance Figure 3. Very similar to the model for expected ground covered, we see features like the linebacker's x and y coordinate, the box player's position and the offense's run tendencies pop up as important for determining what the expected bite distance should be for a player on a play (Figure 7). As Figure 4 previewed, a quarterback's dropback depth post-play action was one of the more important variables as well. Figure 8 includes a pre- and post-snap example of two players from a 2017 game and their respective BDUE estimates.

4 Analysis of metrics for individual players

4.1 Ground covered over expected (GCOE)

One of the primary objectives in developing player-level metrics, such as GCOE and BDUE, is to create measures that are more stable and therefore predictive of perfor-

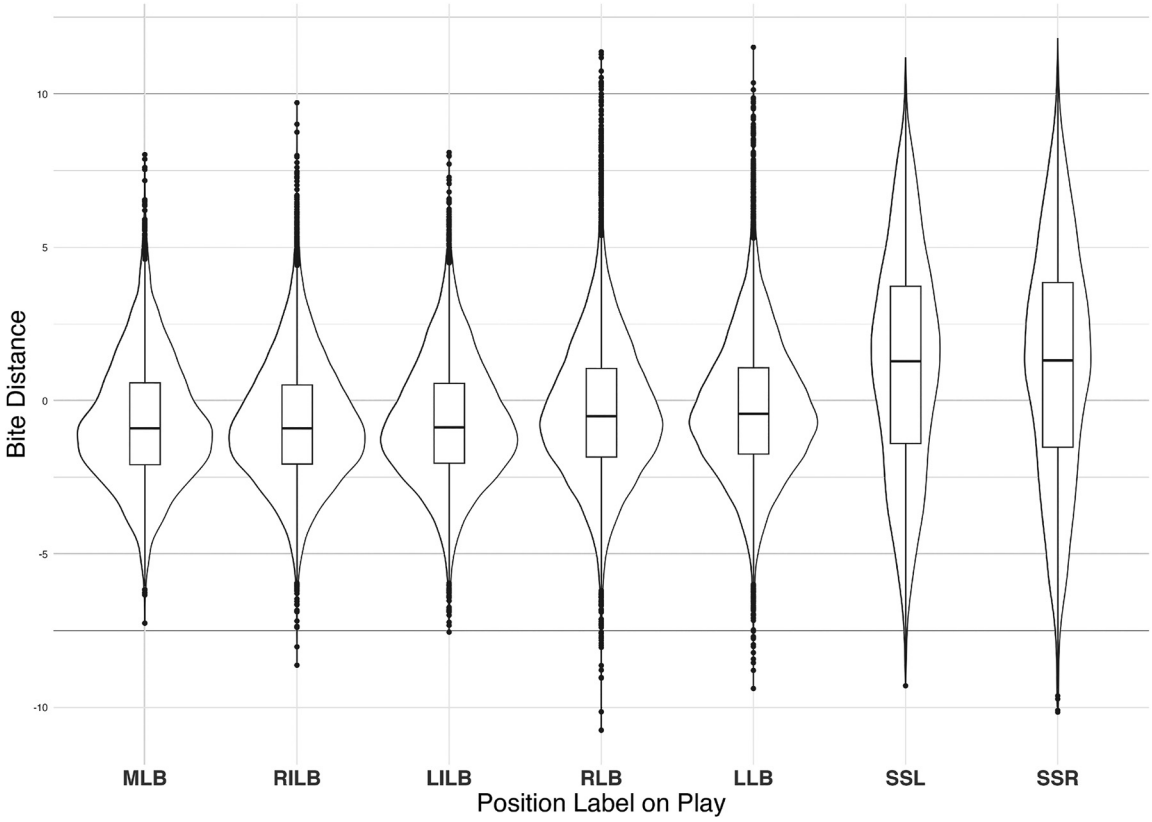


Figure 5: Bite distance by box players as a function of position they played at the snap of the ball.

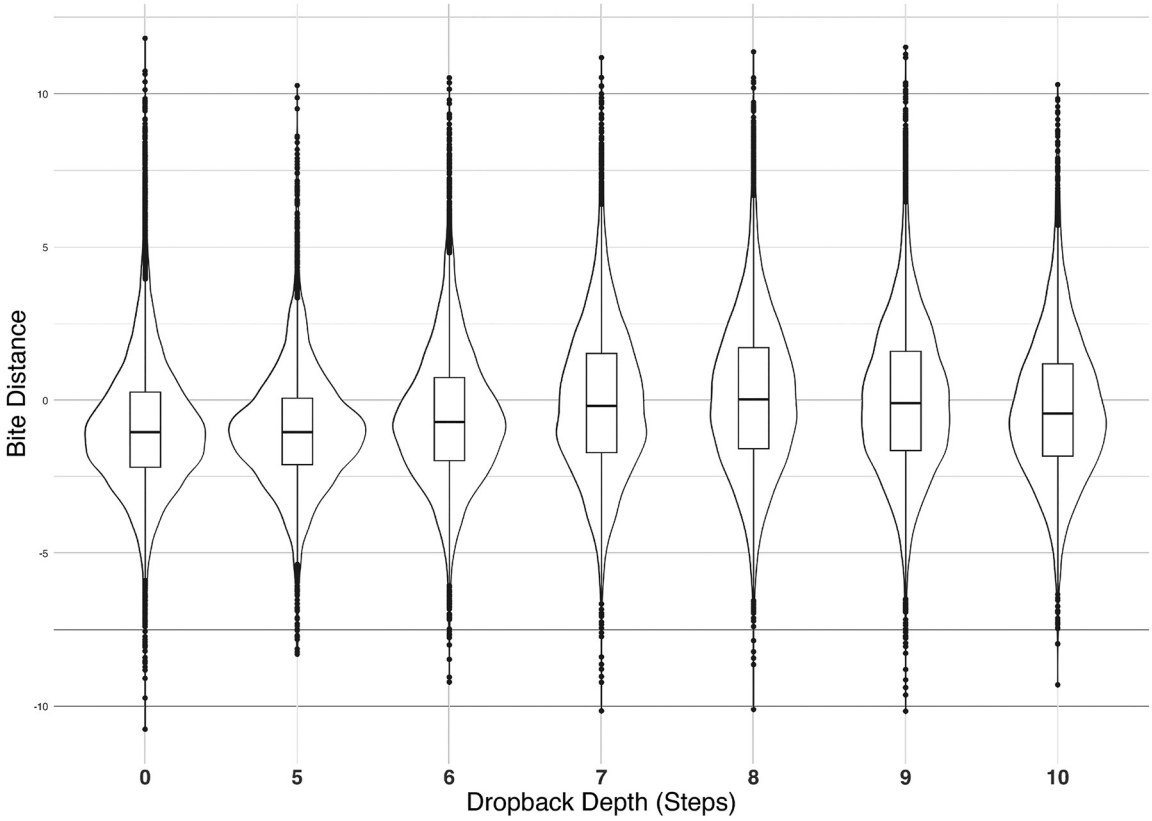


Figure 6: Bite distance by box players as a function of the quarterback's dropback depth on the play-action play.

Table 2: RMSE values for different proposed modeling frameworks for BDUE.

Model	RMSE
Null model	2.96 yards
Elastic-net regression	2.34 yards
Random forest	2.23 yards
XGBoost tuned	2.27 yards

mance from one season to the next. In contrast to coverage data, which can be volatile and subject to variability based on opposing offensive strategies, certain metrics derived from PFF data, such as player grades and "run stops", demonstrate greater year-to-year stability relative to other American football metrics.

To analyze run-defense-related-metrics, we focus our attention on players charted as off-ball linebackers (LLB/RLB, LILB/RILB, or MLB) or strong safeties (SSL/SSR, SS) by PFF and have recorded 200 or more such run defense snaps during the 2017–2021 seasons against rushers classified as running backs or quarterbacks. Our primary metric of interest is run-stop percentage, which measures the rate of run stops per run defense snap. We find that run-stop percentage is correlated year-to-year at a rate of $r = 0.41$ (Figure 9a), while raw +/− PFF grade (the average of the play-level grades assigned by PFF graders) is correlated at a rate of $r = 0.46$ (Figure 9b).

Despite both of those metrics being good proxies for run defense stability, GCOE is even more stable with an r value of $r = 0.66$ (Figure 10) for the 136 players that played

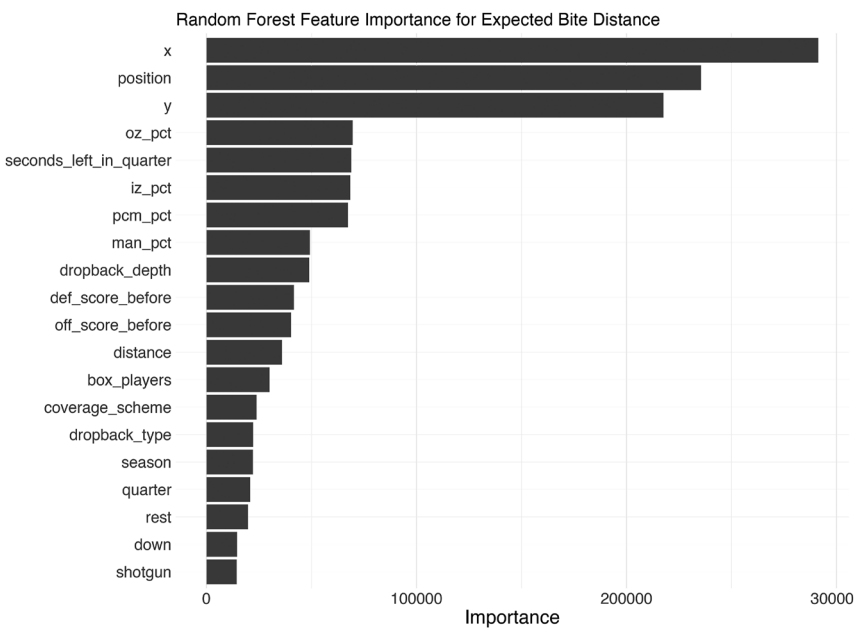


Figure 7: Variable importances for final BDUE model. Only the top 20 features (of 27) are shown.

(a) (b)

Figure 8: Example ground bit distance under expected instances from a 2017 game between Baltimore and New Cincinnati. The figure (a) is pre-snap and (b) is post snap.

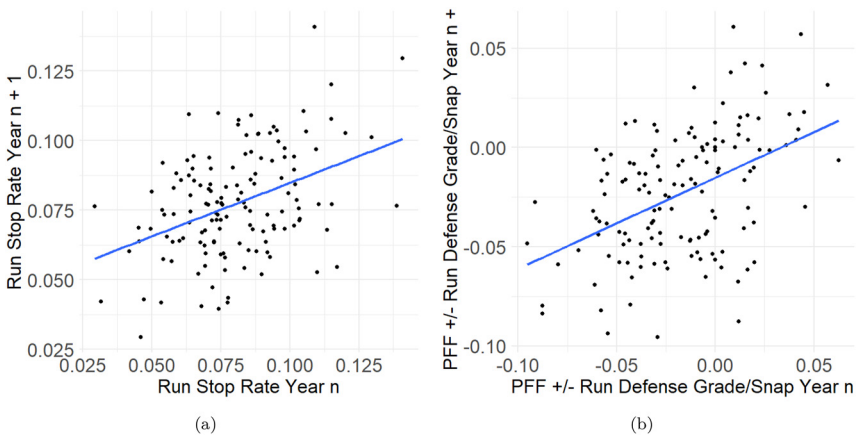


Figure 9: Year-to-year stability for PFF charting data, with run-stop rate (a) and PFF +/− grade per snap (b). Minimum 200 run-defense snaps in a box position.

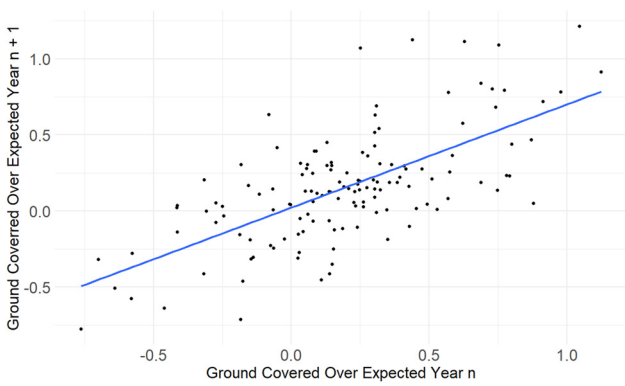


Figure 10: Year-to-year relationship between ground covered over expected (GCOE) and itself. Minimum 200 run-defense snaps in a box position.

200 or more run defense snaps in both seasons. GCOE is also correlated with run stop and PFF grade metrics the following year, as shown in Table 3.

In a linear model for run stop rate in year $n+1$, the inclusion of run stop rate in year n results in an r-squared value of 0.17. When GCOE in year n is also included in the model, the r-squared value increases to 0.20. Similarly, for PFF +/− grade per run defense snap in year $n+1$, the r-squared value is 0.22 when only PFF +/− grade in year n is used as a predictor. With the inclusion of GCOE in year n in the model increases the r-squared value to 0.23.

4.2 Bite distance under expected (BDUE)

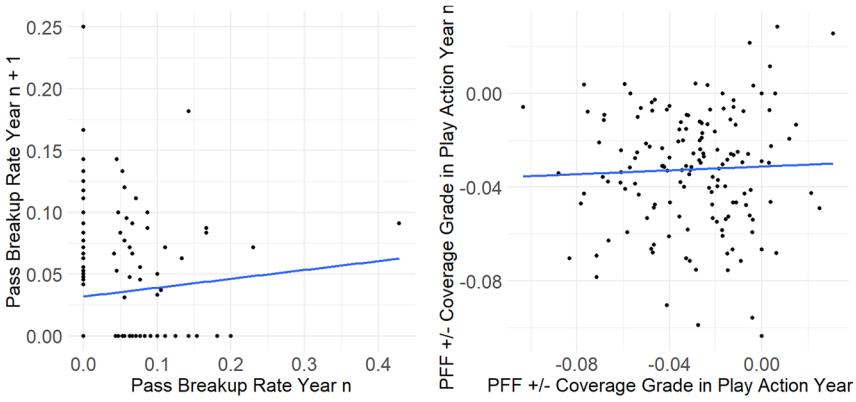
Bite Distance Under Expected (BDUE) was already shown in Eager et al. (2022) to be both a descriptive and predictive alternative to existing coverage metrics for linebackers,

Table 3: Year-to-year relationship between ground covered over expected (GCOE) and PFF charting metrics. Minimum 200 run-defense snaps in a box position.

Metric	Correlation with GCOE n	Correlation with GCOE $n-1$
Stop rate	0.26	0.33
PFF +/− grade	0.24	0.20
GCOE	1	0.66

which are often unreliable from one year to the next (Figure 11). This is particularly the case in low-sample events, such as on play-action passes. For instance, pass breakup rate, which is the frequency with which a player is designated as a primary or secondary coverage player on an incomplete pass by PFF, has a correlation coefficient of $r = 0.09$ from one year to the next for players who have faced 75 or more play-action snaps within a season (Figure 11a). PFF coverage grades on such plays are correlated at a an even lower rate (0.04, Figure 11b).

BDUE exhibits greater stability than the aforementioned coverage metrics, as evidenced in Table 4, correlating at a rate of $r = 0.57$ year to year (Figure 12). Moreover, BDUE outperforms PFF +/− coverage grade against play action in terms of predicting itself from one season to the next. Specifically, a linear regression model that includes PFF +/− coverage grade and BDUE in year n yields a regression coefficient for PFF +/− grade that is not statistically significant at the 0.05 level, indicating that BDUE possesses all the predictive power. This is an advantageous development for BDUE, as it addresses a situation where PFF charting data inadequately captures box players' performance when facing opposition play action.



(a) (b)

Figure 11: Year-to-year stabilty for PFF charting data, with pass-breakup rate (a) and PFF +/− grade per play-action snap (b). Minimum 75 play-action snaps against in a box position.

Table 4: Year-to-year relationship between bite distance under expected (BDUE) and PFF charting metrics. Minimum 75 play-action snaps faced in a box position.

Metric	Correlation with BDUE n	Correlation with BDUE $n-1$
Pass breakup rate	0.02	0.06
PFF +/− grade	0.03	0.15
BDUE	1	0.57

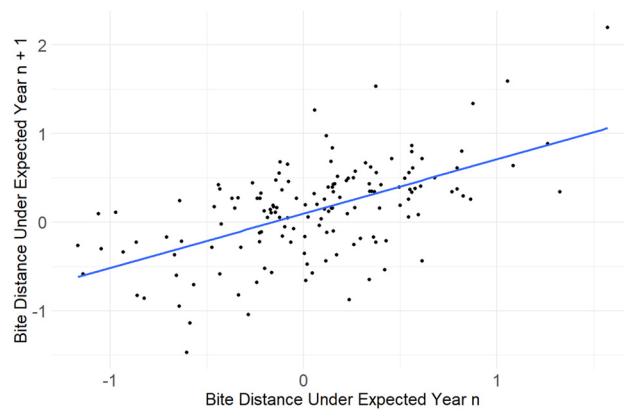


Figure 12: Year-to-year relationship of bite distance under expected (BDUE). Minimum 75 play-action snaps faced in a box position.

4.3 The relationship between GCOE and BDUE

In Sections 4.1 and 4.2, we showed that both GCOE and BDUE improve our understanding of box-player traits by measuring how well a player flows to the ball on run plays while also showing how well they see through an offense's run fake on a play-action pass. These metrics, moreover, are related – they illustrate the trade-off between flowing to the ball in the run game and biting too hard on play action. Players can either be aggressive in nature (flowing well in the run game but biting hard on play-action) or conservative (not flowing downhill to the ball on run plays but not biting on play-action). The correlation between GCOE and BDUE for players who met the snap thresholds ($n = 287$) in the same season was -0.40. This makes intuitive sense, as the more one flows in the run game, the more they also bite on play-action (Figure 13).

In addition to correlating with each other in a given season, we see some predictive power for GCOE in BDUE (Table 5). Other model choices (e.g. elastic-net regressions,

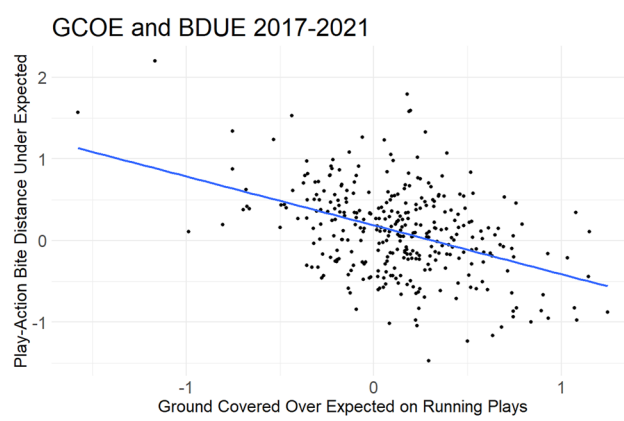


Figure 13: Relationship within year between GCOE and BDUE. Minimum 75 coverage snaps against play action and 200 run defense snaps played at a box position.

Table 5: Year-to-year relationship between GCOE and BDUE. Minimum 75 play-action snaps and 200 run-defense snaps played in a box position. There were 122 players who met both thresholds in adjacent seasons from 2017 to 2021.

Metric	Correlation with BDUE $n+1$	Correlation with GCOE $n+1$
BDUE	0.57	−0.40
GCOE	−0.40	0.66

and even the null model) produced very similar year-to-year correlations both within and between BDUE and GCOE.

4.4 Leaderboards

With the stability of our metrics established, we compare our implied player rankings to consensus opinion. Table 6 displays the league's best players according to BDUE over the course of the 2017–2021 seasons.

The list features some of the most accomplished linebackers of the current NFL era, including several who are likely to earn a place in the Pro Football Hall of Fame. Luke Kuechly, for instance, is a strong candidate for enshrinement, while Dont'a Hightower has a good chance as well. Jamie Collins, K.J. Wright, Eric Kendricks, and Fred Warner all secured lucrative contracts based on their outstanding play during their rookie contracts. In addition, Deone Bucannon, Jordyn Brooks, and Devin White were all highly regarded prospects and first-round picks in the NFL Draft, known for their athleticism and talent. White, in particular, is an intriguing case, as he played a key role in leading the Tampa Bay Buccaneers' defense to victory in the 2020 Super Bowl, despite receiving a low PFF grade. This illustrates the difference between this metric and PFF grades.

Table 7 presents the top-performing players according to GCOE during the 2017–2021 seasons. This group is not as distinguished as the leaders in BDUE, which is consistent with the declining importance of the run game in the modern NFL.

Lastly, taking an average of these two metrics we can uncover the best all-around linebackers by a combination of these two measures in Table 8. Even though coverage is considered a more important kill than stopping the run, both metrics are weighed evenly for simplicity.

The list comprises some of the top players in the linebacker position in the league, including Kuechly, Hightower, and Collins, all of whom were previously mentioned. Additionally, there are other players who are highly

Table 6: Career BDUE values for linebackers from 2017 to 2021. Minimum 150 play-action snaps played in a box position.

Player	Team	n	BDUE
Dont'a Hightower	NE	236	1.21
Thomas Davis Sr.	CAR/LAC/WAS	323	0.921
Jamie Collins	CLE/NE/DET	316	0.845
Devin White	TB	445	0.807
Denzel Perryman	LAC/LV	350	0.705
Neville Hewitt	MIA/NYJ/HOU	324	0.705
Jordyn Brooks	SEA	225	0.678
Luke Kuechly	CAR	423	0.663
Eric Kendricks	MIN	763	0.625
Deon Bucannon	ARZ/NYG/TB	181	0.557
Fred Warner	SF	699	0.542
K.J. Wright	SEA/LV	371	0.520

Table 7: Career GCOE values for linebackers from 2017 to 2021. Minimum 400 run-defense snaps played in a box position.

Player	Team	n	GCOE
Raekwon McMillian	MIA/LV	719	1.01
Zach Cunningham	HOU/TEN	1583	0.834
Kiko Alonso	MIA/NO	832	0.759
Denzel Perryman	LAC/LV	882	0.647
Ja'Whaun Bentley	NE	772	0.627
Duke Riley	ATL/PHI/MIA	533	0.601
Germaine Pratt	CIN	794	0.600
James Burgess	CLE/NYG/GB	422	0.590
Sean Lee	DAL	609	0.586
Jerome Baker	MIA	972	0.570
Kenneth Murray	LAC	402	0.566
Jarrad Davis	DET/NYJ	1130	0.542

Table 8: Career composite BDUE/GCOE values for linebackers from 2017 to 2021. Minimum 150 play-action and 400 run-defense snaps played in a box position.

Player	Team	BDUE	GCOE
Dont'a Hightower	NE	1.209	0.273
Denzel Perryman	LAC/LV	0.705	0.647
Luke Kuechly	CAR	0.663	0.268
Sean Lee	DAL	0.295	0.586
Jamie Collins	CLE/NE/DET	0.845	0.017
Jerome Baker	MIA	0.248	0.570
Jordyn Brooks	SEA	0.678	0.133
C.J. Mosley	BAL/NYJ	0.464	0.319
Zach Brown	WAS/PHI	0.380	0.353
Danny Trevathan	DEN/CHI	0.427	0.258
K.J. Wright	SEA/LV	0.520	0.163
Ja'Whuan Bentley	NE	0.013	0.627

regarded in the league, such as C.J. Mosley, who earned a high-value free-agent contract from the Jets before the 2019 season. Long-term starters in the league, such as Denzel Perryman, Sean Lee, and K.J. Wright, are also on the list. Jerome Baker's contract with the Miami Dolphins made him one of the top-10 paid players at the linebacker position at the time of signing.

5 Analysis of metrics for opposing teams

In addition to evaluating box players, we also analyze how opposing offenses influence linebacker movement.

5.1 GCOE against

We examine the offenses for which opposing box players flow the most and the least against the run, and consider the maximum GCOE value among box players on a given run play. While run defense, like pass defense, is a weak-link system (where outcomes depend on the play of the unit's worst players), the linebacker position is more strong link (depending on the strongest players) than the defensive line, making this seem like the optimal approach. Table 9 shows the top five offenses in getting the opposing defense to flow to the run.

There are some interesting teams in here. Firstly, the first four teams in the list all have running backs that have signed top-of-market deals recently, Green Bay's Aaron Jones, Carolina's Christian McCaffrey, Tennessee's Derrick Henry and New Orleans' Alvin Kamara. While there's a decent amount of evidence that individual value at the running back position is less than historical consensus, this would provide evidence that running backs have value (Sze Yui 2021). Two LaFleur brothers, Green Bay's Matt and New York's Mike, appear here, giving some credence to the idea that style of offense causes box players to flow heavily against run plays.

The five best teams in terms of curbing flow from linebackers on actual running plays are shown in Table 10. In this list you have three athletic quarterbacks in Kansas City's Patrick Mahomes, Jacksonville's Trevor Lawrence and Arizona's Kyler Murray. Linebackers not wanting to be out of position, and thus being less reactive to "eye candy" like play action, makes some sense. What's interesting about the Steelers being on this list is that their quarterback, Ben Roethlisberger, was a notoriously awful play-action passer, as he was the second-worst, worst, and second-worst in yards per attempt on such plays during his final three years as a starter, per PFF (PFF). A good question here is a chicken or egg question; do linebackers move slowly against Pittsburgh because Roethlisberger is not a threat on play action, or does their movement cause play action not to be not effective?

The San Francisco 49ers' offensive play caller, Kyle Shanahan, has long been lauded for his ability to put linebackers in conflict, to the point where it's very likely that they are not moving as much at all anymore as a precautionary measure. To that point, linebackers against the 49ers flowed at the seventh-highest rate in Shanahan's first year (2017), and eighth-highest in his second year (2018), before dropping to the sixth-lowest in 2019 and third-lowest in 2020.

5.2 BDUE against

To determine how much or little opposing defenses bite on play action, we use the average minimum BDUE for an opposing defense's box players during the 2021 season. We used the minimum value because the number of box players on each play varies, making an average value inappropriate. Furthermore, since defense is considered a weak-link system, the performance of the weakest link is crucial to the outcome (Eager 2020). Table 11 displays the top five teams in 2021 for eliciting high negative BDUE.

Here, we see the teams that employed three of the top running backs in football in 2021: Tennessee (Derrick Henry), Cleveland (Nick Chubb) and Indianapolis (NFL rushing leader Jonathan Taylor). Two poor offenses, Detroit and Houston, round out the list. Neither team had much in the

Table 9: Highest average maximum ground covered by opposing box players on run plays during the 2021 season.

Season	Team	Max GCOE against
2021	Green Bay Packers	1.69
2021	Carolina Panthers	1.62
2021	Tennessee Titans	1.62
2021	New Orleans Saints	1.62
2021	New York Jets	1.60

Table 10: Lowest average maximum ground covered by opposing box players on run plays during the 2021 season.

Season	Team	Max GCOE against
2021	Kansas City Chiefs	0.879
2021	Jacksonville Jaguars	0.924
2021	San Francisco 49ers	0.938
2021	Arizona Cardinals	0.971
2021	Pittsburgh Steelers	0.977

Table 11: Lowest (e.g. most negative) average minimum bite distances against by box players on play-action passes during the 2021 season.

Season	Team	Min BDUE against
2021	Tennessee Titans	−1.24
2021	Cleveland Browns	−1.20
2021	Indianapolis Colts	−1.18
2021	Detroit Lions	−1.17
2021	Houston Texans	−1.13

way of a passing offense in 2021, so defenses could sell out to stop the run against them, and hence the highly-negative BDUE values against.

Table 12 looks at the bottom five teams in eliciting BDUE. Notice that, much like for GCOE, the Kansas City Chiefs, with one of the best play callers of all time in Andy Reid and one of the best quarterbacks in all of football in Patrick Mahomes, guide an offense against which linebackers simply don't move. The Chiefs are among the leaders in both of these for the entire NGS era (2017-present). Buffalo quarterback Josh Allen is also an athletic quarterback, whose mobility perhaps causing box players to move less, like what we saw in Table 10. Figure 14 shows the relationship between GCOE and BDUE against for the 2021 season for the entire NFL.

Table 12: Highest (e.g. least negative) average minimum bite distances against by box players on play-action passes during the 2021 season.

Season	Team	Min BDUE against
2021	Washington Football Team	−0.436
2021	Kansas City Chiefs	−0.569
2021	Buffalo Bills	−0.605
2021	Tampa Bay Buccaneers	−0.672
2021	Las Vegas Raiders	−0.684

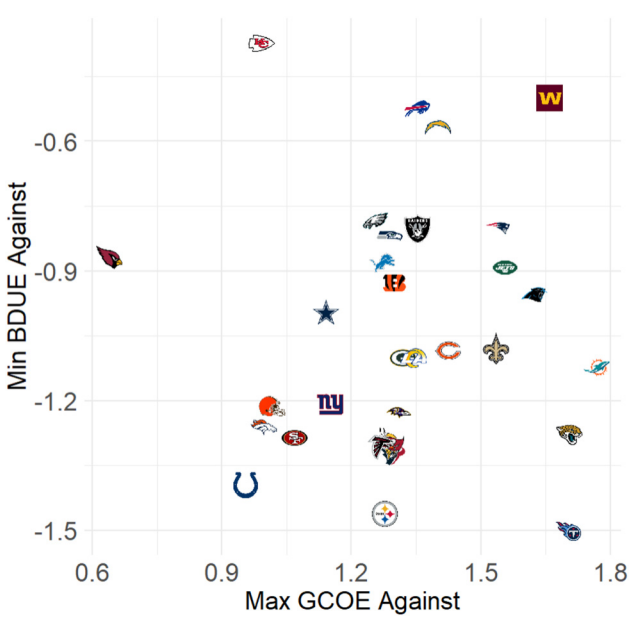


Figure 14: Relationship between GCOE and BDUE against for NFL offenses.

6 Discussion

In this paper, we build on Eager et al. (2022) to refine and develop novel metrics for understanding linebacker instincts in the National Football League. While significant progress has been made in the evaluation of American football players by researchers both in the public and private sphere, measures of linebacker play, especially in coverage, have not necessarily aligned with how the players are valued in the NFL marketplace. Players like Fred Warner, Jerome Baker, Devin White and others are considered near-replacement-level or below by grading companies like PFF at times but are cherished by the league. Some of this perception is reclaimed in this paper, as the aforementioned players are highly rated by our metrics. Additionally, typical coverage metrics for linebackers haven't shown year-to-year stability in the past, but BDUE shows that there is opportunity for stable linebacker metrics involving coverage.

Using (x, y) tracking data we have made significant progress in this work in terms of codifying "traits" who measurements are usually reserved for the scouting department (Bliss 2022; Eager et al. 2022). A linebacker through these metrics can earn an "aggressive" label or a "timid" one based on where they sit in Figure 14. Tracking data provides a repeatable, consistent and objective means to measure these traits with data, which significantly aids in the process of evaluation. Once this tracking data becomes ubiquitous in the college sphere, projecting linebackers from college to the NFL will be more sophisticated and robust than in years past.

Understanding how space is occupied and manipulated is one of the biggest promises stemming from available tracking data, as was highlighted in this work. Teams like the Kansas City Chiefs and San Francisco 49ers, both of which have made multiple conference championships in the 2017–2021 seasons, show that their offenses manipulate offenses well.

Further work in the tracking data sphere should aim to derive traits for players that are stable and map to outcomes that we care about in football. For example, in

Eager et al. (2022) we looked at pass-rush get off rates in an effort to model athleticism of edge players. This metric correlated significantly with things like 40-yard dash, vertical and broad jump. Similar studies for defensive backs would add a great deal to the public understanding of those positions, as up until this point the state-of-the-industry metrics are not particularly stable or predictive.

Acknowledgment: The authors would like to thank SumerSports and PFF for their support during this research, and the Sloan Sports Analytics Conference Research Paper Competition for their feedback on an early version of the BDUE metric and for funding for Tej Seth's trip to the conference in 2022. The authors would also like to thank the two anonymous reviewers, the editors, and associated editors for their constructive feedback on the initial submission, as well as Shawn Syed for his comments on the writing of the manuscript.
Author contributions: All the authors have accepted responsibility for the entire content of this submitted manuscript and approved submission.
Research funding: None declared.
Conflict of interest statement: The authors declare no conflicts of interest regarding this article.

References

Baldwin, B. 2018a. *Rushing Success and Play-Action Passing*. https://www.footballoutsiders.com/stat-analysis/2018/rushing-success-and-play-action-passing.
Baldwin, B. 2018b. *Further Research on Play-Action Passing*. https://www.footballoutsiders.com/stat-analysis/2018/further-research-play-action-passing.
Baldwin, B. 2022. *nflfastR, EP, WP, CP, xYAC, and xPass Models*. https://www.opensourcefootball.com/posts/2020-09-28-nflfastr-ep-wp-and-cp-models/.
Bliss, T. 2022. *Pass Rushing Edge Get Off Speed*. https://www.kaggle.com/code/tombliss/pass-rushing-edge-get-off-speed.
Burke, B. 2013. *How Coaches and the NYT Bot Compare*. https://www.nytimes.com/newsgraphics/2013/11/28/fourth-downs/post.html.
Burke, B. 2019. "DeepQB: Deep Learning with Player Tracking to Quantify Quarterback Decision-Making & Performance." In *2019 Sloan Sports Analytics Conference*. https://www.sloansportsconference.com/research-papers/deepqb-deep-learning-with-player-tracking-to-quantify-quarterback-decision-making-performance.
Burriel, B., and J. M. Buldu. 2021. *The Quest for the Right Pass: Quantifying Player's Decision Making*. https://statsbomb.com/2021/11/statsbomb-conference-2021-research-papers/.
Chu, D., M. Reyers, J. Thomason, and L. Y. Wu. 2020. "Route Identification in the National Football League: An Application of Model-Based Curve Clustering Using the EM Algorithm." *Journal of Quantitative Analysis in Sports* 16 (2): 121—32.
Clark, K. 2019. *The Race to Make the NFL an Exact Science*. https://www.theringer.com/nfl/2019/4/22/18510099/nfl-draft-analytics-player-tracking.
Deshpande, S., and K. Evans. 2020. "Expected Hypothetical Completion Probability." *Journal of Quantitative Analysis in Sports* 16: 85—94.
Eager, E. 2020. *How Fragile is Defense: A Look at the Effect of Stars versus Depth on Game Outcomes*. https://www.pff.com/news/nfl-how-fragile-is-defense-a-look-ateffect-of-stars-versus-depth-on-game-outcomes.
Eager, E., and G. Chahrouri. 2019. *PFF Data Study: Coverage vs. Pass Rush*. https://www.pff.com/news/pro-pff-data-study-coverage-vs-pass-rush.
Eager, E., B. Brown, G. Chahrouri, T. Riske, B. Spielberger, L. Sze Yui, Z. Drapkin, and T. Seth. 2022. "Using Tracking and Charting Data to Beter Evaluate NFL Players: A Review." In *2022 MIT Sloan Sports Analytics Conference*. https://www.sloansportsconference.com/research-papers/using-tracking-and-charting-data-to-better-evaluate-nfl-players-a-review.
Football Outsiders. 2019. *Play-Action Passing and Game Conditions*. https://www.footballoutsiders.com/stat-analysis/2019/play-action-passing-and-game-conditions.
Football Outsiders. 2022. https://footballoutsiders.com.
Hasan, A. 2002. *Minnesota Vikings: How Fullback C.J. Ham is the Key to Understanding the New-Look Offense*. https://theathletic.com/3572885/2022/09/07/minnesota-vikings-cj-ham-offense/?redirected=1.
Hermsmeyer, J. 2019. *Can NFL Coaches Overuse Play-Action? They Haven't Yet*. https://fivethirtyeight.com/features/can-nfl-coaches-overuse-play-action-they-havent-yet/.
Lee, D. 2021. *Lee: A Brief Dialogue About RPOs in the NFL*. https://www.pff.com/news/nfl-dialogue-rpos-miami-dolphins-2021.
Life Magazine. 1969, November 21st Issue.
Lopez, M. 2020. "Bigger Data, Better Questions, and the Return to Fourth Down Behavior: An Introduction to a Special Issue on Tracking Data in the National Football League." *Journal of Quantitative Analysis in Sports* 16 (2): 73—9.
Next Gen Stats. 2022. https://nextgenstats.nfl.com/.
Palazzolo, S. 2018. *Breaking Down PFF's Run Concepts*. https://www.pff.com/news/breaking-down-pffs-run-concepts.
PFF. 2022. https:://pff.com.
R Core Team. 2021. *R: A Language and Environment for Statistical Computing*. Vienna: R Foundation for Statistical Computing.
Reyers, M., and T. B. Swartz. 2023. "Quarterback Evaluation in the National Football League Using Tracking Data." *AStA Advances in Statistical Analysis* 107: 327—42.
Rodrigue, J. 2021. *Inside the Rams' Major Changes to Their Draft Process, and Why They Won't Go Back to 'normal'*. https://theathletic.c,om/2507342/2021/04/12/inside-the-rams-major-changes-to-their-draft-process-and-why-they-wont-go-back-to-normal/.
Sze Yui, L. 2021. *Running Backs Don't Matter: A Dive Back into the Popular NFL Analytics Phrase*. https://www.pff.com/news/nfl-running-backs-dont-matter-a-dive-back-into-the-popular-nfl-analytics-phrase.
XGBoost. 2022. https://xgboost.readthedocs.io/en/latest/.